At sea, fish have become victim to the kind of overexploitation once reserved for animals on land. Plummeting stocks of once-common species, such as tuna and cod, are typical of a resource that is often only weakly regulated, or not at all. Some fish breed at an early age, and can recover from overfishing if the pressure is reduced. But with species like tuna, maturity takes time, so adult fish can become too rare to guarantee a future supply of young. Comparatively little is known about the effect of this relentless harvesting on marine and coastal life. However, fish play a key part in many food chains, and when their numbers fall, the effects are felt by countless other animals, from seabed invertebrates to fish-eating birds.

Introduced species

Even before Columbus discovered America, explorers and colonists had spread animals to new parts of the world. The process increased rapidly with the Age of Exploration, and the result—hundreds of years later—is that the wildlife of isolated regions has been overwhelmed by a host of intruders, from rats and cats to mosquitoes. Some of these introduced species cause problems by actively preying on local wildlife. Others harm native animals indirectly, by competing with them for food, or by transmitting diseases such as avian malaria.

In Australia, introduced species have disrupted the ecology of an entire continent. Kangaroos still thrive, but many small marsupials now live in a tiny fraction of their original range, in marginal habitats that introduced species find difficult to reach. Similar problems affect New Zealand and Madagascar, and on much smaller oceanic islands, the situation is often more severe. Their native birds are often wiped out by cats and rats—tenacious newcomers that are extremely difficult to eradicate. In this age of rapid travel and expanding tourism, the threat from introduced species is never far away.

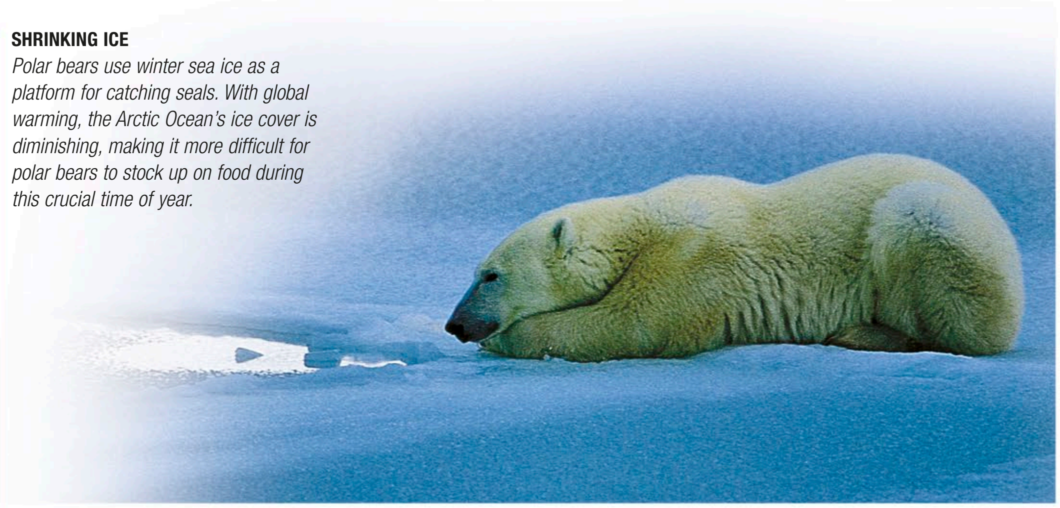
UNWELCOME ARRIVALS
Rabbits were introduced to Australia in the mid-19th century for food and their fur. They quickly spread inland, displacing native animals and destroying vegetation. In dry areas, soil erosion set in, permanently changing the landscape.

Global warming

The world's climate has always changed, but the current period of rapid warming is without precedent in modern times. Most scientists believe that the cause is increasing levels of atmospheric greenhouse gases, caused by human activities. Greenhouse gases include water vapor, chlorofluorocarbons (CFCs), and carbon dioxide. They make the atmosphere trap outgoing heat, warming up the earth.

Wildlife has coped with changes in the past, but the speed and severity of this episode could result in extinction on a worldwide scale. Atmospheric warming, in itself, is only part of the story, because global warming has dozens of knock-on effects. It makes sea ice melt and changes the pattern of oceanic currents, altering climatic conditions on land. Over the longer term, it makes the oceans more acid, threatening shelled animals and coral reefs. It also makes sea levels rise, both by melting ice caps, and by making seawater expand. This expansion happens very slowly, but once started, will take centuries to reverse.

The United Nations Framework Convention on Climate Change negotiated the so-called Paris Agreement for cutting greenhouse gas emissions. It entered into force in November 2016 and has, to date, been ratified by 132 parties. Many see this as the critical turning point in reducing global warming.



SHRINKING ICE
Polar bears use winter sea ice as a platform for catching seals. With global warming, the Arctic Ocean's ice cover is diminishing, making it more difficult for polar bears to stock up on food during this crucial time of year.

Animals on the brink

The International Union for Conservation of Nature (IUCN) maintains The IUCN Red List of Threatened Species™ (see panel, below), which is constantly updated by the work of scientists worldwide. In 2017, nearly 76,000 species were assessed, representing about 3 percent of those that have been formally classified. Although this is only a small portion of the world's species, this sample indicates how life on earth is faring, how little is known, and how urgent the need is to assess more species. Comprehensive assessments have been carried out for birds, mammals, amphibians, sharks, reef-building coral, cycads, and conifers, and the statistics are disturbing, with 1 in 8 birds, 1 in 4 mammals, 1 in 3 corals, and more than 1 in 3 amphibians at risk of extinction.

Being "on the brink" means different things for different species. Some animals—particularly invertebrates—can reproduce rapidly when conditions are good, which means that they have the potential to make a fast comeback. But many species on the IUCN Red List are slow breeders, and take a long time to recover if their numbers fall. Albatrosses are typical examples: they take up to 7 years to become mature, lay just one egg, and often breed only in alternate years.

To make matters more complex, animals cannot necessarily breed if they find a suitable habitat, and a partner of the opposite sex. This is because many species breed in groups, and rely on the stimulus of others around them to trigger essential behavior, such as courtship and nest-building. The passenger pigeon was a classic example of a communal breeder, nesting in colonies many square miles in extent. Even when many thousands were left, it had already stepped over the threshold into oblivion.

EXTINCT IN THE WILD: SCIMITAR-HORNED ORYX

CRITICALLY ENDANGERED: BLACK RHINOCEROS

ENDANGERED: QUEEN ALEXANDRA'S BIRDWING

VULNERABLE: WANDERING ALBATROSS

NEAR THREATENED: RED-EARED SLIDER

LEAST CONCERN: RUFOUS BETTONG

THREAT CATEGORIES
The IUCN Red List of Threatened Species places animals in one of 8 categories according to the degree of risk they face: the most threatened species (such as the orangutan and the black rhinoceros) are "critically endangered"; the next category (including the Queen Alexandra's birdwing) are "endangered," and so on. Details of the categories—which are also used in this book—can be found on page 10.

THE IUCN RED LIST

The IUCN Red List of Threatened Species is published by the International Union for Conservation of Nature (IUCN). The IUCN, founded in 1948 by the United Nations, carries out a range of activities aimed at safeguarding the natural world. Part of its work is the regular compilation of the IUCN Red List, which draws together information provided by over 10,000 scientists from all over the world; this list has become a global directory to the state of living things on our planet.

The current IUCN Red List shows that threatened species are often grouped in particular parts of the world. Today's "hot spots" include East Africa, Southeast Asia, and the American tropics. One of the reasons for this is that these regions have a much greater diversity of species than regions farther north or south: the American tropics, for example, are particularly rich in bird species. In recent years, these areas have seen rapid habitat change—particularly deforestation—which has come about partly because an expanding human population needs more land on which to grow food.